

Lab 1A: String Induction

The following problems ask you to prove some “obvious” claims about recursively-defined string functions. In each case, we want a self-contained, step-by-step induction proof that builds on formal definitions and prior results, *not* on intuition. In particular, your proofs must refer to the formal recursive definitions of string length and string concatenation:

$$|w| := \begin{cases} 0 & \text{if } w = \varepsilon \\ 1 + |x| & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

$$w \bullet z := \begin{cases} z & \text{if } w = \varepsilon \\ a \cdot (x \bullet z) & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

The **reversal** w^R of a string w is defined recursively as follows

$$w^R := \begin{cases} \varepsilon & \text{if } w = \varepsilon \\ x^R \bullet a & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

For example, $\text{STRESSED}^R = \text{DESSERTS}$ and $\text{WTF374}^R = 473FTW$.

1. Prove that $|w| = |w^R|$ for every string w .

Base case: Suppose $w = \varepsilon$

Then,

$$\begin{aligned} |w^R| &= |\varepsilon^R| \\ &= |\varepsilon| \quad \text{by definition of reversal} \\ &= |w| \end{aligned}$$

2.

3.